

A TUTORIAL REPORT

21CSE423T - BIG DATA VISUALIZATION

SUBMITTED BY

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BDA-A IV

1. BUILDING INTERACTIVITY IN PLOTS

Interactive plots enable users to explore data dynamically through hovering, zooming, panning, filtering, and tooltips. Interactive visualizations are widely used in educational analytics, business intelligence, and scientific research.

Objectives

- To understand interactive visualization concepts.
- To create plots with hover information.
- To enable zooming and panning features.
- To analyze online course enrollment trends.

Tools Required

Python, Pandas, Plotly, Jupyter Notebook

Methodology

Step 1: Import Libraries

```
import pandas as pd
```

```
import plotly.express as px
```

Step 2: Create Dataset containing course names and enrollment counts.

Step 3: Create an interactive Plotly bar chart showing enrollment distribution across courses.

Dataset and Code

```
data = {  
    "Course":["Python","Data Science","Machine Learning","Cloud Computing","Cyber Security"],  
    "Enrollments":[450,380,320,290,210]}
```

```
fig = px.bar(df,x="Course",y="Enrollments",title="Online Course Enrollment  
Analysis",color="Course")  
fig.show()
```

Features of Interactivity

- Displays exact enrollment values on hover.
- Allows zooming into specific regions.
- Supports panning and exploration.

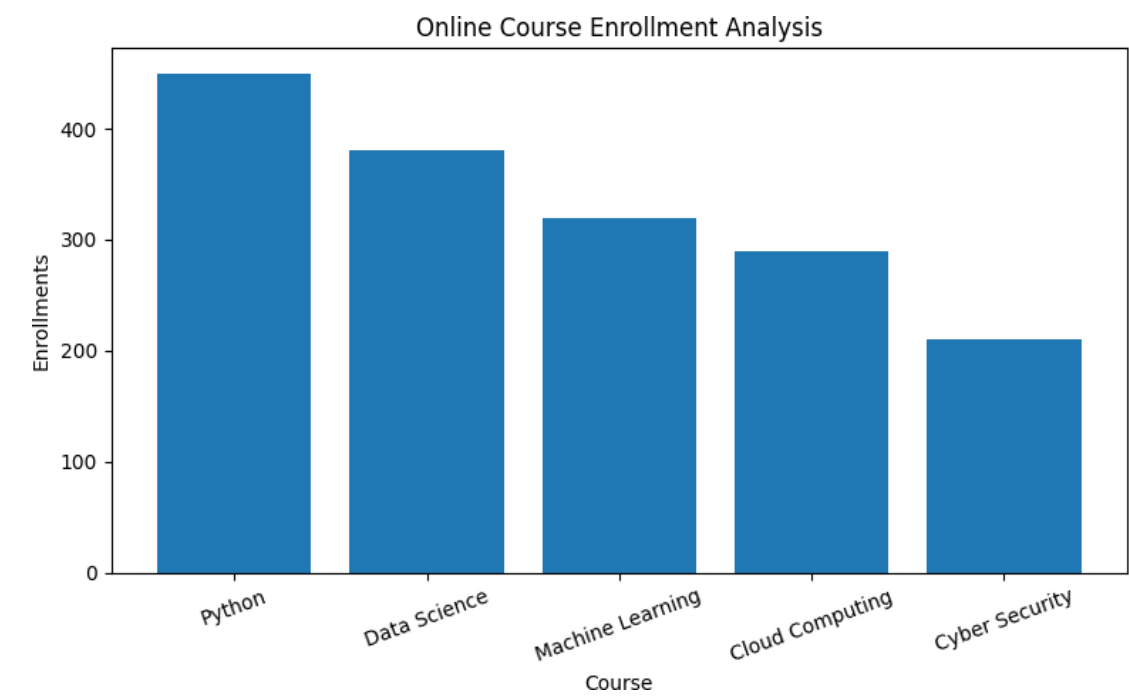
Observations

- Python course has the highest enrollments.
- Cyber Security has the lowest enrollments.
- Hover tooltips provide exact values.
- Interactive features improve data exploration.

Advantages, Applications and Conclusion

Interactive plots improve user engagement, support trend analysis, and enhance decision-making. Applications include educational analytics, online learning platforms, business intelligence, and research. Interactive visualizations simplify interpretation of large datasets.

OUTPUT



2. BUILDING DASHBOARD USING PANEL

A dashboard combines multiple visualizations and controls into a single interface. Panel is an open-source Python library that supports interactive dashboards using Plotly, Bokeh, Matplotlib, and Pandas.

Objectives

- To create an interactive dashboard.
- To display student performance metrics.
- To provide user-friendly controls for analysis.
- To understand dashboard development using Panel.

Tools Required

Python, Pandas, Plotly, Panel

Methodology

- Step 1: Install Panel using pip.
- Step 2: Import Panel, Pandas and Plotly.
- Step 3: Create a student performance dataset.
- Step 4: Generate a Plotly visualization.
- Step 5: Embed the chart inside a Panel dashboard.

Dataset and Code

```
data = {  
    "Subject":["Python","DBMS","DAA","Machine Learning","Cloud"],  
    "AverageMarks":[86,79,82,91,88]}  
  
dashboard = pn.Column("# Student Performance Dashboard", pn.pane.Plotly(fig))  
dashboard.show()
```

Dashboard Components

Header – Displays dashboard title.
Charts – Shows subject-wise average marks.
Interactive Controls – Supports user interaction.
Layout Management – Organizes components neatly.

Observations

- Machine Learning has the highest average marks.
- DBMS has the lowest average marks.
- Dashboard presents academic information clearly.
- Interactive charts improve understanding.

Advantages, Applications and Conclusion

Panel enables rapid dashboard creation with responsive layouts and widget support. Applications include academic analytics, student monitoring, institutional reporting, and business dashboards. Dashboards improve accessibility and decision-making.

OUTPUT

